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Quotation Management System

SECTION : 03

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1.0 OVERVIEW OF THE PROJECT	7
2.0 PROBLEM STATEMENT	9
<i>As a result, the current system at DOSTTO:</i>	10
3.0 PROPOSED SOLUTIONS	11
3.1 User Registration and Login	11
3.2 Quotation Management Module	11
3.3 Report Management Module	12
3.4 Feedback Module	12
3.5 User Management Module	12
3.6 Data Storage and Security	12
3.7 Benefits of the Proposed System	13
Summary	13
4.0 CURRENT BUSINESS PROCESS / WORKFLOW	14
4.1 Step-by-Step Business Workflow (AS-IS)	14
Step 1: Customer Request	14
Step 2: Employee Prepares the Quotation	14
Step 3: Manager Reviews the Quotation	14
Step 4: Sending Quotation to Customer	14
Step 5: File and Record Keeping	15
Step 6: Report Generation	15
4.2 Problems in the Current Workflow	15
4.3 Example Scenario	15

5.0 LOGICAL DFD (AS-IS)	16
5.1 Context Diagram (AS-IS)	16
External Entities:	16
Data Flows:	16
Main Process:	16
Diagram: Context Diagram (AS-IS)	17
5.2 Diagram 0 (AS-IS System)	18
Internal Processes:	18
Data Stores (Manual):	18
External Entities (same as before):	18
Diagram: Level 0 DFD (AS-IS)	18
5.3 Child Diagram (AS-IS – for Process 2: Prepare Quotation)	19
Sub-Processes:	19
Data Stores Used:	19
Diagram: Child DFD – Process 2 (AS-IS)	19
5.4 Summary of the AS-IS System	20

6.0 SYSTEM ANALYSIS AND SPECIFICATION (TO-BE)	21
6.1 Logical DFD (TO-BE System)	21
6.1.1 Context Diagram (TO-BE)	22
6.1.2 Level 0 DFD (TO-BE)	23
6.1.3 Child Diagram for Process 2: Prepare Quotation	24
6.2 Process Specification (TO-BE)	25
Process 1: Receive Quotation Request	25
Process 2: Prepare Quotation	25
Process 3: Approve Quotation	25
Process 4: Send Final Quotation	25
Process 5: Generate Reports	26
Summary	26

7. PHYSICAL SYSTEM DESIGN	27
7.a Hardware Requirements	27
For Customers (external users).....	27
For Employees (internal users).....	27
For Managers and Admins	27
For Server (optional if cloud-based)	27
7.b Software Requirements	28
Client Side (for users).....	28
Server Side	28
Development Tools	28
7.c Network and Communication	29
Setup	29
Communication.....	29
7.d System Access and Roles	29
7.e System Features Distribution.....	29
7.f Advantages of the New Physical Design	30
Conclusion	30

7.1 PHYSICAL DFD TO-BE SYSTEM	31
<i>Physical DFD (Diagram 0 – TO-BE)</i>	<i>31</i>
<i>Child Diagram – Prepare Quotation (Physical).....</i>	<i>32</i>
<i>Partitioning</i>	<i>32</i>
<i>CRUD Matrix (Create, Read, Update, Delete)</i>	<i>33</i>
<i>Event Response Table.....</i>	<i>33</i>
<i>Structure Chart.....</i>	<i>34</i>
<i>System Architecture</i>	<i>35</i>
Architecture Layers:	35
Deployment.....	35
<i>Summary</i>	<i>35</i>
 8.0 SYSTEM WIREFRAME (INPUT/OUTPUT DESIGN).....	36
8.1 Input Design	36
1. Customer Quotation Request Form	36
2. Employee Quotation Entry Form.....	37
3. Manager Review Panel.....	37
4. Admin Report Filter.....	37
8.2 Output Design	38
1. Customer Confirmation Screen	38
2. Employee Dashboard	38
3. Manager Dashboard	38
4. Final Quotation (PDF Output).....	39
5. Reports Module Output	39
8.3 Wireframe Examples (Text Format)	40
A. Customer Request Form (Wireframe)	40
B. Manager Approval Screen (Wireframe)	41
C. Admin Report Screen (Wireframe)	42
Summary	42

9.0 SUMMARY OF THE PROPOSED SYSTEM	43
<i>Purpose of the System</i>	<i>43</i>
<i>Who Will Use the System</i>	<i>43</i>
<i>Key Features of the System.....</i>	<i>44</i>
<i>Benefits of the System</i>	<i>44</i>
1. Time Saving.....	44
2. Improved Accuracy.....	44
3. Centralized Data.....	44
4. Better Communication	44
5. Security and Control.....	44
6. Detailed Reports	45
7. Customer Satisfaction.....	45
<i>System Technology and Architecture</i>	<i>45</i>
<i>How It Solves the Existing Problems</i>	<i>45</i>
<i>Future Scalability.....</i>	<i>46</i>
<i>Final Summary.....</i>	<i>46</i>

1.0 Overview of the Project

This project is a team assignment from two important university subjects: **System Analysis and Design** and **Database Systems**. These two subjects are part of the Information Technology program. The purpose of this project is to help students learn how to design a real system for a real company. It teaches both theory (the knowledge) and practice (the skills).

In this project, students work as if they are system analysts in a real company. They talk to a real client, try to understand the client's problems, and then design a system that can help fix those problems. The client for this project is a company called **DOSTTO**.

DOSTTO is a company that works in many areas:

- **Housing** (helping people find or build homes),
- **Education** (training and learning services),
- **Oil Refining** (fuel and energy),
- **Industry** (factories and production services).

Right now, DOSTTO is facing problems in how it manages its employees and data. Most of their work is done **manually** — using papers, Excel files, and email. There is no central system. Because of this, they have many problems like:

- Wasting time on small tasks,
- Losing files,
- Making errors in reports,
- Repeating the same work,
- And not knowing how well their staff are working.

The goal of this project is to **design a new computer system** that can help DOSTTO work better and faster. This new system is called a **Quotation Management System**. It will help DOSTTO:

- Receive customer quotation requests online,
- Let employees prepare the quotations easily,
- Allow managers to approve quotations quickly,
- And allow admins to view all activities and reports in one place.

In the **System Analysis and Design** part of the project, students learn how to build the system step-by-step. They do the early stages of system development. This includes:

- Studying the current system,
- Finding problems in the current way of working,
- Talking to users and gathering information,
- Writing down what the system should do (requirements),
- Creating diagrams to show how the new system will work.

They follow the **Software Development Life Cycle (SDLC)**. This is a process used in real companies to plan and design systems before building them.

In the **Database Systems** part of the project, students learn how to design the **backend** of the system. That means the part that stores all the data. They follow the **Database System Development Life Cycle (DBSDLC)**. They do:

- Conceptual design (making an idea of the database),
- Logical design (planning tables, relationships, and data types),
- Normalization (removing data duplication),
- Planning how the system will store and use data using SQL.

By combining both courses, this project gives students a **complete view of system development**. It shows how to design a real system from the beginning, with proper planning and teamwork. The result of the project is a **fully designed system** that is not only good for learning but can also be used in a real company.

This system will help DOSTTO solve many of its current problems. It will:

- Make quotation work faster and easier,
- Keep data in one safe place,
- Improve how managers track employees,
- Provide useful reports to help make better decisions.

2.0 Problem Statement

At DOSTTO, the way employee tasks and internal activities are managed is still **fully manual**. There is **no proper digital system** to handle employee information or daily work. Right now, everything is done using **paper files**, **Excel spreadsheets**, or **verbal messages**. This old system causes many problems for the company.

For example:

- **Attendance** is written in notebooks or Excel.
- **Task assignments** are often given by word of mouth or on chat apps.
- **Work progress** is hard to follow.
- **Reports** are made by hand and take a long time.

This way of working is **slow**, **confusing**, and often leads to **mistakes**.

There is **no central place** where all data is stored. Some data is saved on computers, some on paper, and some in emails. Because of this, it is **hard for managers** to find the latest and correct information. Many departments work on their own, so there is **no clear communication** between them.

Here are more problems DOSTTO is facing:

- **Performance tracking** is difficult. There is no tool to see how much work an employee has done.
- **Scheduling and leave approval** is slow and done manually.
- **Duplicate data** is common. The same data is sometimes entered in two or three different places.
- **Reports are delayed** because employees have to collect and check data manually.
- **Security is weak**. Many employees can access shared files, so no one knows who changed what. Sensitive documents may be stored in unlocked rooms or shared folders.
- **Data loss** is a big risk. Files on personal computers can be deleted by mistake or damaged by viruses. There is no backup system.

Managers also have no automatic way to compare:

- Tasks assigned to each employee,
- Tasks completed,
- Department performance over time.

This lack of control makes it very hard for leaders to make smart decisions.

As a result, the current system at DOSTTO:

- Makes office work slow and heavy,
- Causes confusion and errors,
- Allows too many people to access private data,
- Has no proper tracking or reporting tools,
- And stops the company from growing or improving.

DOSTTO needs a better system to manage employee work, track performance, and keep all information in one safe, easy-to-use place.

3.0 Proposed Solutions

To solve the problems in DOSTTO's current manual system, we propose the development of a new digital system called the **Quotation Management System (QMS)**. This system will help DOSTTO manage quotations, monitor employee activities, and store all important data in one safe place. The new system will replace Excel files, paper forms, and verbal messages with a smart, easy-to-use web-based platform.

The proposed system will support **three main types of users**:

1. **Customers** – People who request quotations for services.
2. **Employees** – Staff who prepare and manage the quotations.
3. **Managers or Admins** – Supervisors who approve quotations, manage users, and generate reports.

This new system will include several useful modules. Each module solves specific problems from the old manual system.

3.1 User Registration and Login

- Customers can create an account using their email and basic information.
- Admins will create accounts for employees.
- A secure **login system** will make sure each user can only access the parts of the system they are allowed to.
- The system will support **role-based access** to protect data and privacy.

3.2 Quotation Management Module

This is the **main module** of the system.

- Customers can **request quotations** by filling a simple online form.
- Employees can **prepare the quotation** by entering the required service, prices, and customer details.
- The system will help employees **calculate the total price** and save a **draft quotation**.
- Managers can **approve or reject** the quotation with one click.
- The approved quotation will be **automatically sent to the customer's email**.
- Quotations will be **tracked by status** (e.g., pending, approved, rejected) so that nothing is forgotten or lost.
- Employees can see the list of quotations they are working on.
- Managers can assign quotations to staff and check progress.

3.3 Report Management Module

- The system will automatically create **daily, weekly, and monthly reports**.
- Admins and managers can **filter reports** by employee, department, date, or quotation status.
- This helps managers see:
 - How many quotations were accepted or rejected,
 - Which employees completed the most work,
 - Which departments are most active.
- Reports will help in **making better decisions, planning for the future, and improving performance**.

3.4 Feedback Module

- After receiving the quotation, customers can give **feedback** about the process, pricing, or staff performance.
- Feedback will be stored and grouped by category (positive, negative, suggestion).
- Managers can use this feedback to **improve service quality** or give **staff training**.
- This will also help increase customer satisfaction and build trust.

3.5 User Management Module

- Admins can manage all users in the system.
- They can **add, edit, or remove** employee accounts.
- Admins can assign **roles and permissions** for each user.
- The system will also store logs of all user activity, to help track who did what and when (audit trail).

3.6 Data Storage and Security

- All data will be stored in a **central database**.
- The system will have **backup features** to prevent data loss.
- Passwords and sensitive information will be protected using encryption.
- Only authorized users will be able to see or change important data.
- This will reduce the risk of **data leaks, deletion, or unauthorized access**.

3.7 Benefits of the Proposed System

This system will solve many of the problems found in the current manual system. Some key benefits are:

- **Faster quotation processing** – no more waiting for emails or approvals.
- **Centralized data** – all information is in one place and easy to find.
- **Better security** – no more lost files or shared folders.
- **Clear status tracking** – know what is pending or completed at all times.
- **Employee accountability** – system tracks who did what and when.
- **Easy reporting** – see work performance and quotation summaries in one click.
- **Customer satisfaction** – quicker responses, better service, and a simple way to send feedback.

Summary

The proposed Quotation Management System will help DOSTTO become a more modern and efficient company. It will remove manual work, save time, improve accuracy, and provide useful tools for decision-making. It will also make employees more productive and customers more satisfied.

This solution is designed to grow with the company, so it can be expanded or updated in the future when needed. It is the first step toward DOSTTO's digital transformation.

4.0 Current Business Process / Workflow

This section explains how DOSTTO currently manages its quotation process. The current system is **manual**, meaning that the employees do all tasks by hand. They use tools like **paper forms, Excel spreadsheets, emails, and verbal instructions**. There is **no automatic system** to help manage the work. Because of that, many problems happen daily such as data duplication, lost files, delays, and poor communication.

The quotation process involves **four main people**:

- **Customer** – the person requesting a quotation.
- **Employee** – the staff member who prepares the quotation.
- **Manager** – the person who approves or rejects the quotation.
- **Admin Staff** – the person who saves files, updates records, and creates reports manually.

4.1 Step-by-Step Business Workflow (AS-IS)

Step 1: Customer Request

- A customer wants to get a quotation for a service.
- They contact the company using phone, email, or face-to-face.
- There is no online form or official system for submitting the request.

Step 2: Employee Prepares the Quotation

- The employee receives the request and opens Microsoft Excel or Word.
- They find the customer's old details by searching in emails or saved files.
- The employee enters the required service details, item descriptions, and prices manually.
- The total amount is calculated using a calculator or Excel formula.
- There is no database, so every time, the employee has to start from scratch or copy old files.

Step 3: Manager Reviews the Quotation

- The employee sends the quotation to the manager by email or on paper.
- The manager checks the quotation and gives feedback.
- If changes are needed, the employee edits the file and sends it again.
- If approved, the manager signs it (physically or by email).

Step 4: Sending Quotation to Customer

- Once approved, the employee emails the quotation file to the customer.
- There is no tracking system to check if the customer has seen or responded to the quotation.

Step 5: File and Record Keeping

- The final approved quotation may be saved on the employee's computer or printed and stored in a physical folder.
- Some employees save files with different names, causing confusion.
- There is no shared storage or central system where all quotations are kept.
- Admin staff sometimes have to search through emails or ask other staff to find old files.

Step 6: Report Generation

- When the manager asks for a report (e.g., how many quotations were sent or approved), the admin collects the data manually from emails, Excel sheets, or paper files.
- This takes many hours and is often not accurate.

4.2 Problems in the Current Workflow

1. **No automation** – Every step requires manual work.
2. **Data is scattered** – Information is stored in different formats and locations (emails, USBs, paper, Excel).
3. **No quotation tracking** – There is no way to know if a quotation is pending, approved, or rejected unless someone remembers.
4. **Slow communication** – Feedback from the manager can take days.
5. **File duplication** – Different versions of the same quotation may exist with different names.
6. **Poor reporting** – Reports are not updated or complete.
7. **Lack of accountability** – It's hard to know who worked on what.
8. **High risk of data loss** – If a computer crashes, the files may be gone forever. There is no backup.

4.3 Example Scenario

Here is a real example of how the process works in the current system:

A customer emails DOSTTO asking for a price quotation for a housing project. An employee opens an old Excel file, copies it, and changes the name and details. They print it and leave it on the manager's desk. The manager approves it the next day. The employee then emails it back to the customer. Later, another employee asks for a copy of that quotation, but no one knows where it is saved. The admin staff checks emails and old folders for hours before finding it.

This kind of situation happens often at DOSTTO because the company does not have a proper quotation management system.

5.0 Logical DFD (AS-IS)

A **Data Flow Diagram (DFD)** is a drawing that shows how data moves inside a system. It helps us understand:

- **Who uses the system** (external entities),
- **What processes happen inside** the system,
- **What data is stored** (data stores),
- **What information is sent or received** (data flows).

In this section, we will explain the current system at DOSTTO using DFDs. This system is **manual**, meaning all work is done using paper, Excel, emails, and memory. The system is not computerized, and there is no automation.

5.1 Context Diagram (AS-IS)

The **Context Diagram** is the top-level view of the system. It shows:

- The system as **one big process**,
- All the **users (entities)** who interact with it,
- The **data going in and out**.

External Entities:

- **Customer** – asks for a quotation.
- **Employee** – prepares the quotation.
- **Manager** – approves or rejects the quotation.
- **Admin Staff** – saves files and creates reports.

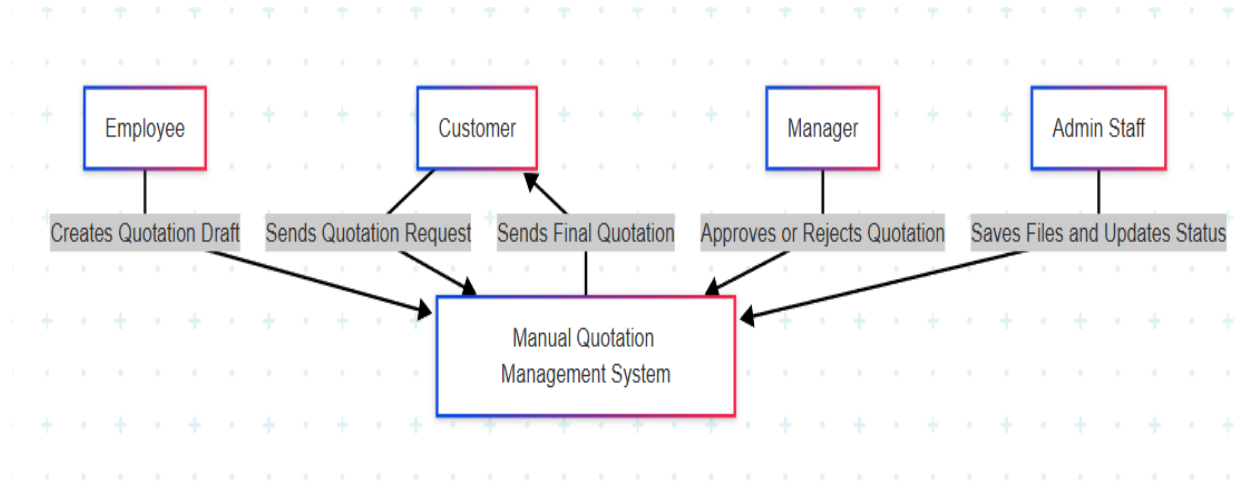
Data Flows:

- Customer sends a **Quotation Request**
- Employee prepares a **Quotation Draft**
- Manager sends **Approval or Rejection**
- System sends **Final Quotation** to Customer
- Admin Staff receives or saves **Quotation Records**

Main Process:

- "Manual Quotation Management System"

Diagram: Context Diagram (AS-IS)



This diagram shows that all users interact with one central process — the manual quotation system — with no real-time tracking or automation.

5.2 Diagram 0 (AS-IS System)

The **Level 0 DFD** breaks the big process into **smaller steps**. Each process represents a task done manually by employees or managers.

Internal Processes:

1. **Receive Quotation Request**
2. **Prepare Quotation**
3. **Manager Reviews Quotation**
4. **Send Final Quotation to Customer**
5. **File and Record the Quotation**

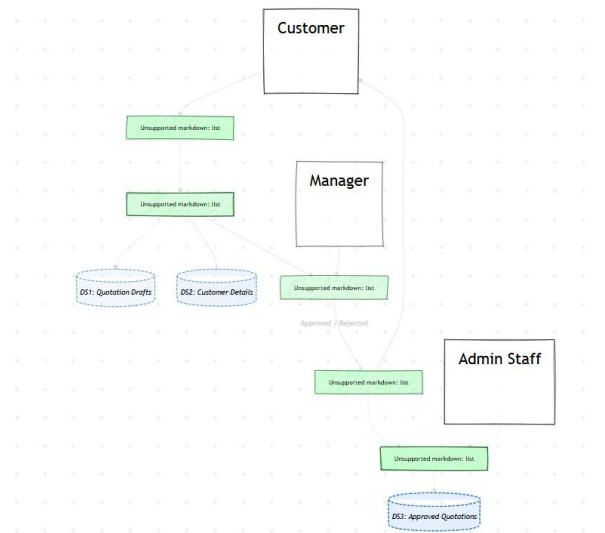
Data Stores (Manual):

- **D1: Quotation Drafts** (saved on PC or printed)
- **D2: Customer Details** (emails, folders)
- **D3: Approved Quotations** (hardcopy or email archive)

External Entities (same as before):

- Customer
- Employee
- Manager
- Admin Staff

Diagram: Level 0 DFD (AS-IS)



This diagram shows the step-by-step manual flow of creating, reviewing, sending, and saving quotation

5.3 Child Diagram (AS-IS – for Process 2: Prepare Quotation)

The process “**Prepare Quotation**” (Process 2 from Diagram 0) has multiple small steps. These are also done manually by employees.

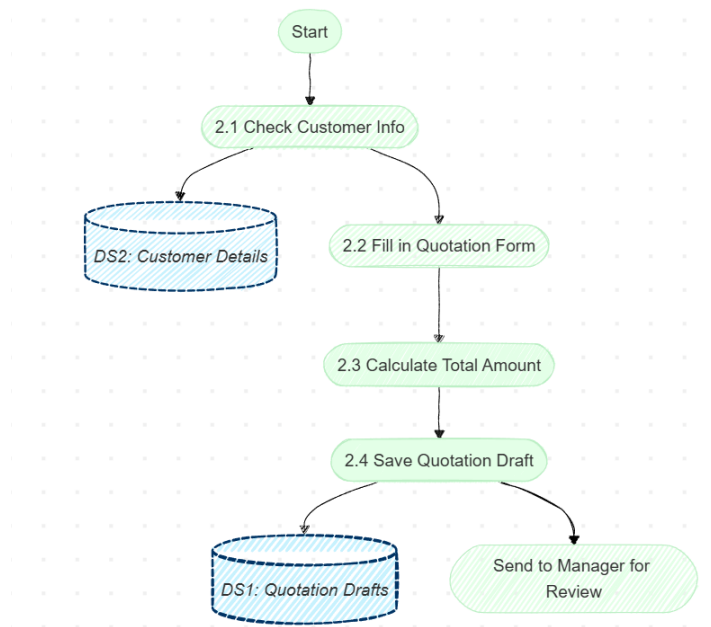
Sub-Processes:

- **2.1 Check Customer Information**
- **2.2 Enter Service Details**
- **2.3 Calculate Total Cost**
- **2.4 Save Draft Quotation**

Data Stores Used:

- Customer Details (D2)
- Quotation Drafts (D1)

Diagram: Child DFD – Process 2 (AS-IS)



This shows how the employee moves through the steps of preparing a quotation.

5.4 Summary of the AS-IS System

- The current system is **manual** and depends on people remembering what to do.
- Data is stored in **different places**, including computers, emails, and paper files.
- There is **no tracking system** to check quotation status.
- Reports are created by collecting data **manually**, which is slow and not accurate.
- Employees must do the same tasks again and again, which wastes time.
- Files can be **easily lost** or saved in the wrong place.
- **No automation**, no reminders, and no performance tools exist in the current system.

This AS-IS DFD analysis shows why DOSTTO needs a new system. The next section will explain the TO-BE system and how it will fix these problems.

6.0 System Analysis and Specification (TO-BE)

This section explains how the **new Quotation Management System (QMS)** will work. It shows the future (TO-BE) version of the system, where all quotation-related tasks are done using a **centralized, automated, and digital platform**.

The TO-BE system is designed to fix the problems in the current manual process. It will improve:

- **Speed of communication**
- **Data security**
- **Tracking and monitoring**
- **File organization**
- **Report generation**
- **Customer service**

This section includes:

- Logical DFD diagrams for the TO-BE system
- Step-by-step explanation of system processes
- Specification of what each process will do

6.1 Logical DFD (TO-BE System)

In this part, we show how the **future digital system** will work using three Data Flow Diagrams:

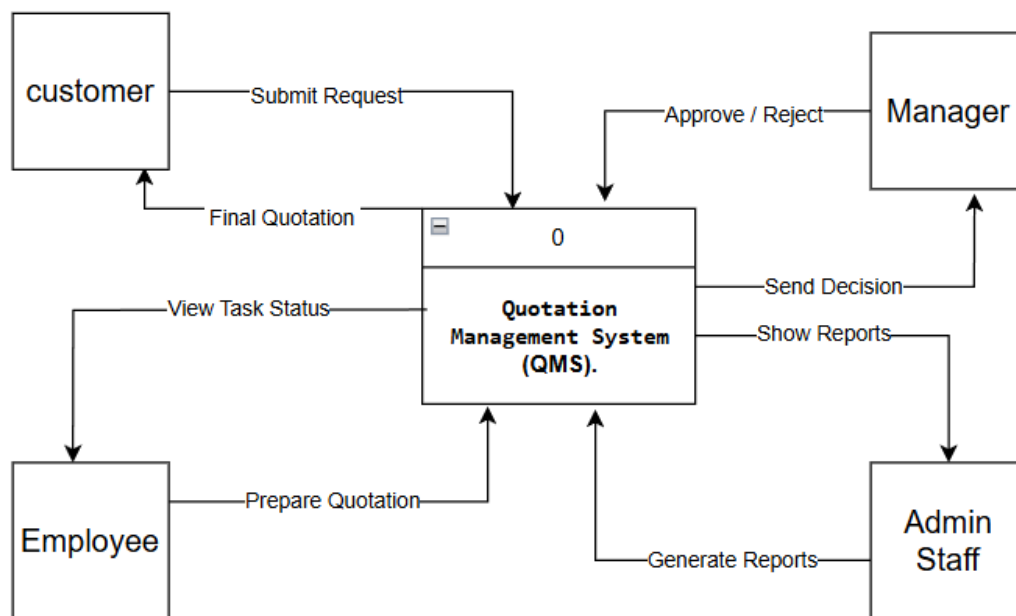
- **6.1.1 Context Diagram**
- **6.1.2 Level 0 DFD**
- **6.1.3 Child Diagram** (for Process 2: Prepare Quotation)

6.1.1 Context Diagram (TO-BE)

This diagram shows a high-level view of how different users interact with the new system.

External Entities:

- **Customer:** Sends quotation request and receives final quotation.
- **Employee:** Prepares quotation.
- **Manager:** Approves or rejects quotation.
- **Admin Staff:** Creates reports and manages users.



6.1.2 Level 0 DFD (TO-BE)

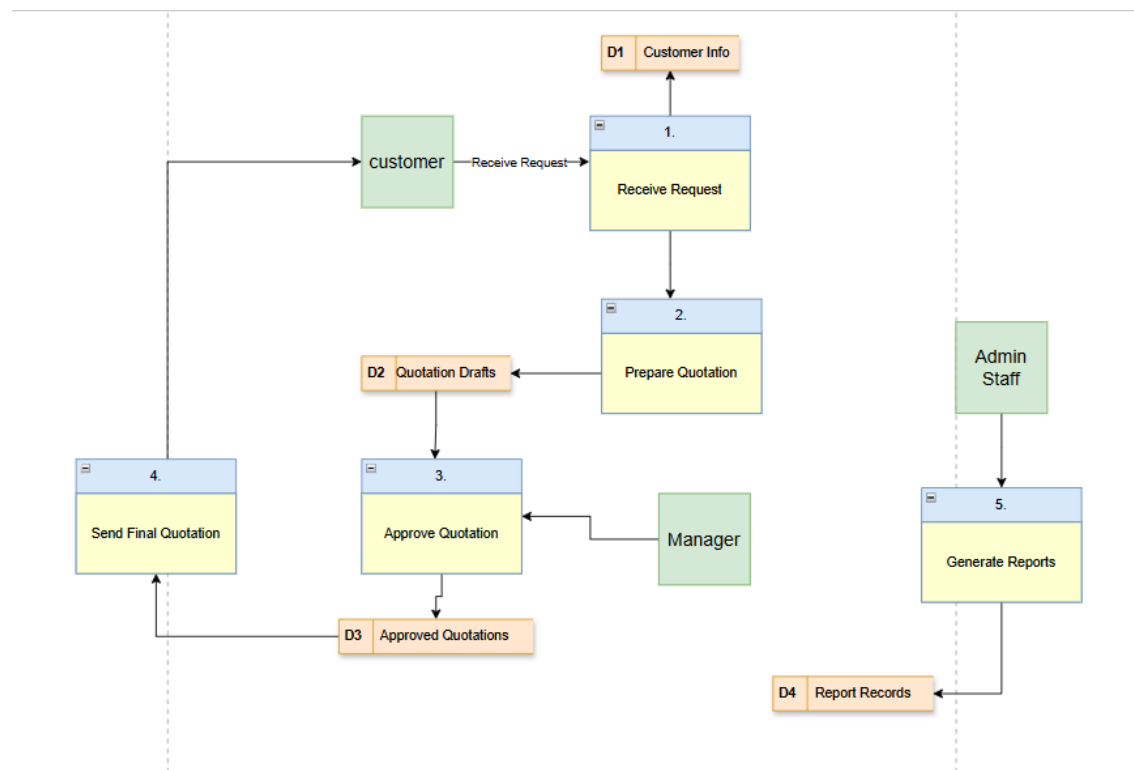
This breaks the system into main processes:

Processes:

1. **Receive Quotation Request**
2. **Prepare Quotation**
3. **Approve Quotation**
4. **Send Final Quotation**
5. **Generate Reports**

Data Stores:

- D1: Customer Info
- D2: Quotation Drafts
- D3: Approved Quotations
- D4: Report Records

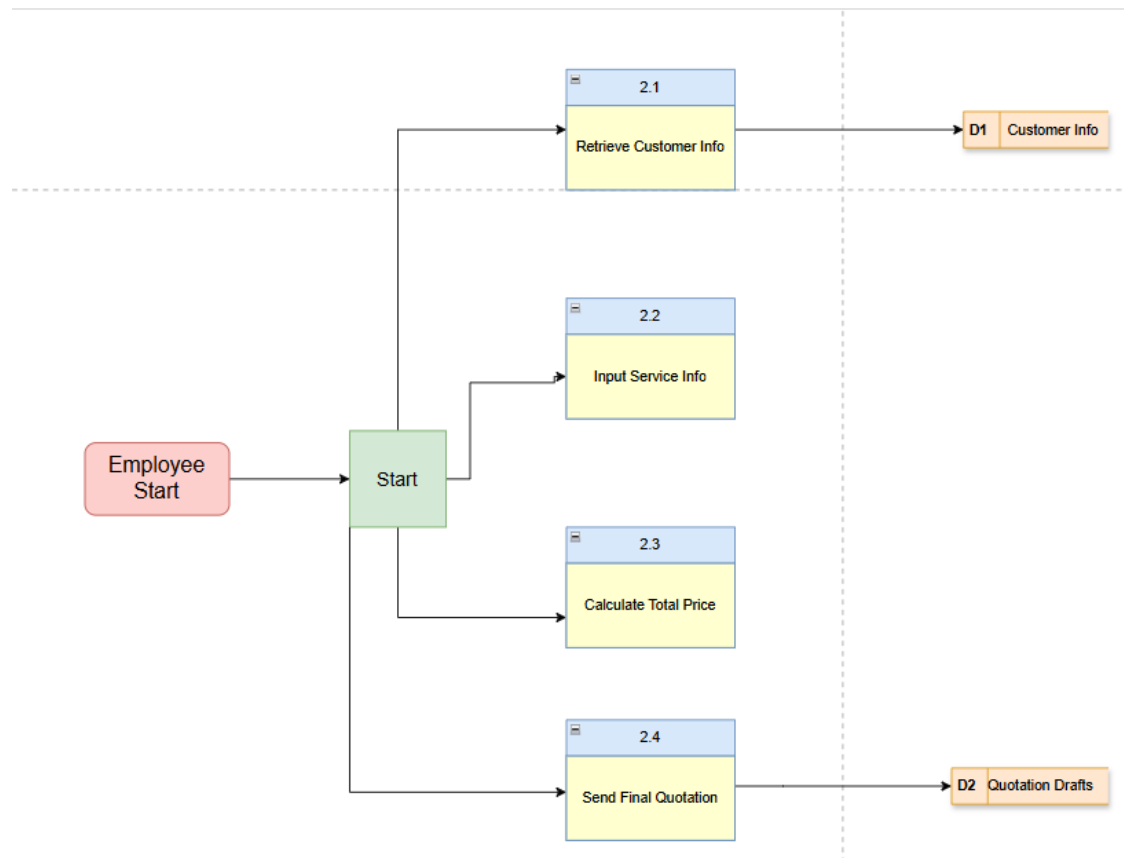


6.1.3 Child Diagram for Process 2: Prepare Quotation

This diagram shows detailed steps done by employees while preparing the quotation.

Steps:

- 2.1: Get customer data from database
- 2.2: Enter service and price
- 2.3: Calculate total cost
- 2.4: Save quotation draft



6.2 Process Specification (TO-BE)

This part explains what each process from Diagram 0 does in simple words.

Process 1: Receive Quotation Request

- **User:** Customer
- **Input:** Name, contact, type of service
- **Function:** System stores request and customer data
- **Output:** Data goes to employee dashboard

Process 2: Prepare Quotation

- **User:** Employee
- **Input:** Customer request
- **Steps:**
 - Search for customer in system
 - Add service description and price
 - Calculate total amount
 - Save draft
- **Output:** Draft stored and marked “Waiting for Review”

Process 3: Approve Quotation

- **User:** Manager
- **Input:** Draft from employee
- **Function:** Manager checks, adds comments, and clicks approve or reject
- **Output:** System updates status

Process 4: Send Final Quotation

- **User:** System
- **Input:** Approved quotation
- **Function:** System sends final quotation to customer’s email
- **Output:** Customer receives PDF copy

Process 5: Generate Reports

- **User:** Admin
- **Input:** Filter options (date, status, employee, department)
- **Function:** System shows statistics like total quotations, response time, approval rate
- **Output:** Report displayed or downloaded as file

Summary

The TO-BE system is **modern, automatic, and easy to use**. It replaces manual tasks with digital features. The system helps all users:

- Customers can request services quickly.
- Employees can manage tasks with less stress.
- Managers can approve and track work faster.
- Admins can generate accurate reports in seconds.

With this system, DOSTTO will be able to work more efficiently and reduce errors and delays.

7. Physical System Design

The physical system design explains the **hardware, software, network setup, and user environment** needed to run the **Quotation Management System (QMS)** at DOSTTO.

The goal of the physical design is to show **how the new system will work in real life**, what tools are needed, and how the system will be used by each user in the company.

7.a Hardware Requirements

This section describes the physical devices (computers, servers, etc.) needed to run the system.

For Customers (external users)

- **Device:** Smartphone, tablet, or personal computer (PC)
- **Internet connection:** Required to access the online system
- **Browser:** Google Chrome, Firefox, Safari, or any modern browser

For Employees (internal users)

- **Device:** Desktop or laptop computer (Windows or Mac)
- **Processor:** Minimum Intel i5 or higher
- **RAM:** 8 GB or more
- **Storage:** At least 250 GB HDD or SSD
- **Network:** Stable internet (LAN or Wi-Fi)

For Managers and Admins

- Same as employees, but with access to full features
- Admin may also require **printer** for downloading reports

For Server (optional if cloud-based)

- **CPU:** Intel Xeon or AMD Ryzen server-grade processor
- **RAM:** 16 GB or more
- **Storage:** 1 TB SSD (for database and files)
- **OS:** Linux (Ubuntu/CentOS) or Windows Server 2019
- **Database Server:** MySQL, PostgreSQL, or SQL Server

Note: The system can also run on a cloud platform like AWS, Google Cloud, or Azure — which means no need for a physical local server.

7.b Software Requirements

Client Side (for users)

- Web Browser
- Operating System (Windows 10+, macOS, Android, iOS)
- PDF viewer (to open final quotations)

Server Side

- **Web Server Software:** Apache or Nginx
- **Database Management System:** MySQL or PostgreSQL
- **Backend Language:** PHP, Python (Django/Flask), or Node.js
- **Frontend Language:** HTML5, CSS3, JavaScript (React or Vue.js)

Development Tools

- Code Editor (e.g., VS Code)
 - Git/GitHub for version control
 - XAMPP/LAMP Stack for local testing
-

7.c Network and Communication

The Quotation Management System will be a **web-based system** that works through the **Internet**. All users will connect to the system using a browser.

Setup

- **Domain:** www.dostto-qms.com (example)
- **Hosting:** Cloud-based server or company-managed web server
- **Security:** HTTPS with SSL certificate

Communication

- **Customer → System:** via secure web form
 - **Employee → System:** via employee login dashboard
 - **Manager → System:** via approval panel
 - **Admin → System:** via admin control panel
-

7.d System Access and Roles

The system will allow different users to log in based on their role:

User Role	Access Level	Devices Used
Customer	Request quotation, view feedback	PC, phone, tablet
Employee	Create and edit quotations	Desktop or laptop
Manager	Approve, reject, comment	Desktop or laptop
Admin	Full access, report, user control	Desktop with admin tools

Each user will only see the parts of the system they need. This protects data and prevents errors.

7.e System Features Distribution

Feature	Automated?	Manual?	Device
Quotation Request	Yes	No	Any (web)
Quotation Preparation	Yes	No	PC/laptop
Manager Approval	Yes	No	PC/laptop
Email to Customer	Yes	No	System
Report Generation	Yes	No	Admin PC
Feedback Submission	Yes	No	Customer phone/PC

7.f Advantages of the New Physical Design

- **Easy Access:** Works from anywhere using a web browser.
- **Low Cost:** No expensive hardware needed.
- **High Security:** Uses encrypted access (HTTPS, login, backups).
- **Central Database:** Keeps all records in one safe place.
- **Scalable:** Can be expanded in the future if more features or users are needed.

Conclusion

The physical system design of the new Quotation Management System will make it easy to use, safe, and accessible from anywhere. It will be built using reliable hardware and modern software. All users — customers, employees, managers, and admins — will be able to do their tasks smoothly using simple devices and web browsers.

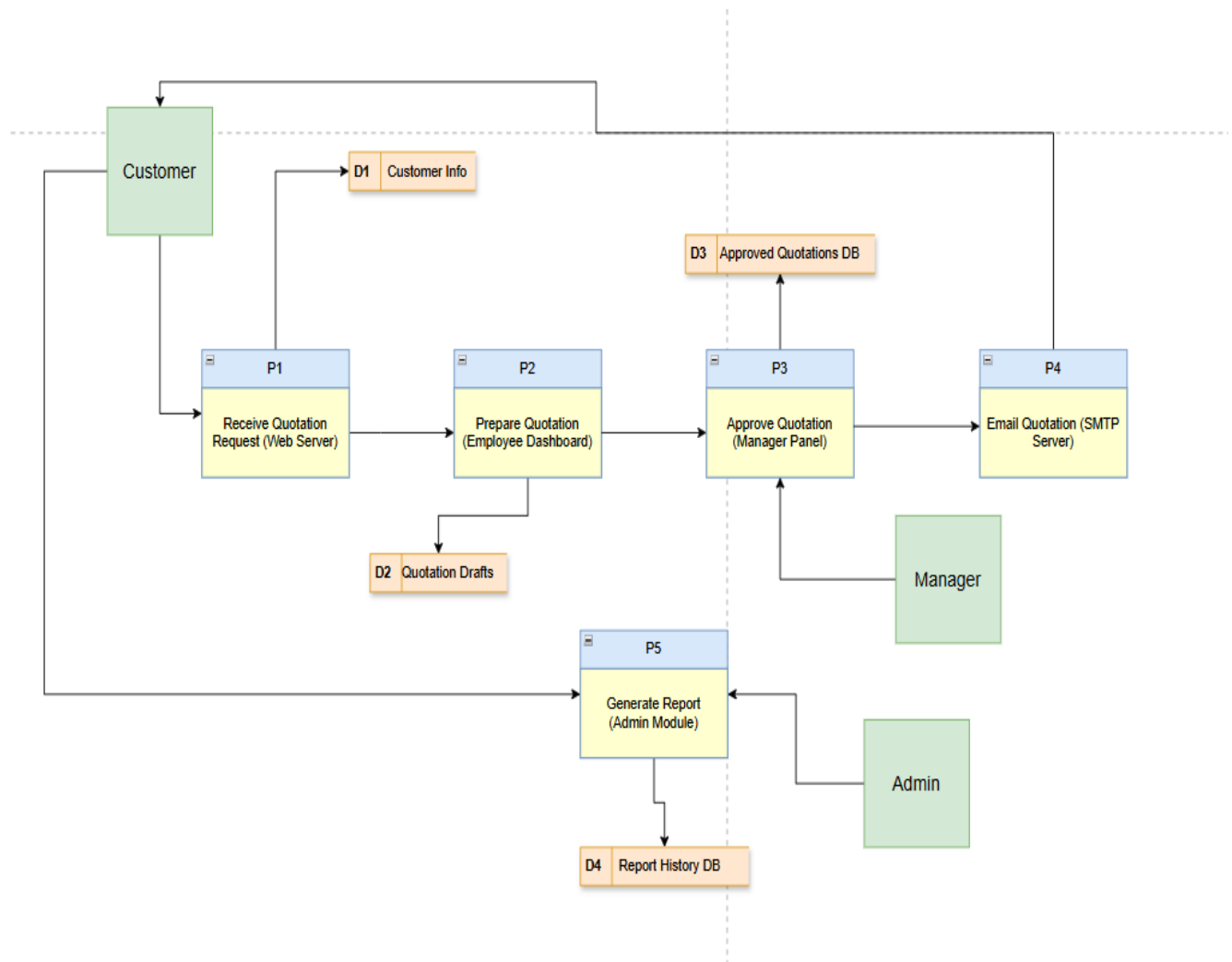
In the next section, we will show the **Physical DFD (TO-BE)** and explain how the system will work at a more technical level.

7.1 Physical DFD TO-BE System

This section explains how the **Quotation Management System (QMS)** will be physically implemented in the real world. It includes all the **technical components, manual vs automated processes**, and the way **data flows between users, processes, and storage**.

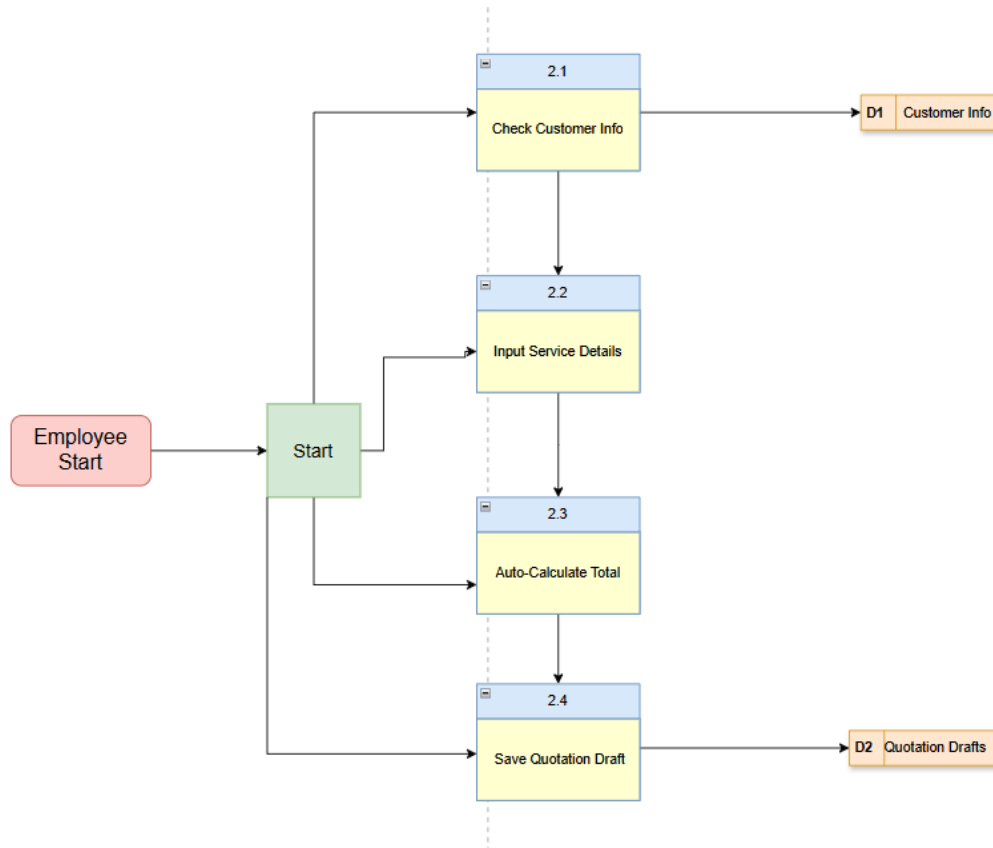
Physical DFD (Diagram 0 – TO-BE)

This diagram shows how the system will work with real components — like computers,



Child Diagram – Prepare Quotation (Physical)

This shows the internal steps taken by an employee when preparing a quotation.



Partitioning

Partitioning explains which parts of the system are done **automatically** (by computer) and which require **human interaction** (manual).

Process	Manual or Automated	Notes
Submit Quotation Request	Manual (by customer)	Done through web form
Store Customer Info	Automated	Saved to DB
Prepare Quotation	Manual (employee)	Filled on dashboard
Calculate Total	Automated	Using backend logic
Manager Approval	Manual	Manager clicks Approve/Reject
Email Final Quotation	Automated	Sent through SMTP
Generate Reports	Automated	Auto-exported based on filter

CRUD Matrix (Create, Read, Update, Delete)

This table shows which user roles can do which actions on each part of the database.

Data Item	Customer	Employee	Manager	Admin
Customer Info	C	R	R	CRUD
Quotation Drafts	-	CRUD	R	R
Approved Quotations	-	R	C	R
Reports	-	-	R	CRUD
User Accounts	-	-	-	CRUD
Feedback	C	R	R	R

Legend:

- C = Create
- R = Read
- U = Update
- D = Delete

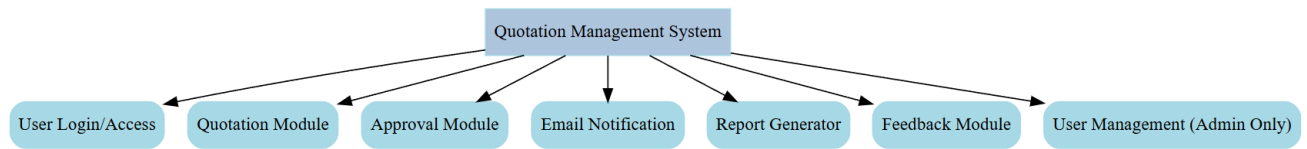
Event Response Table

This table lists the main system events and how the system responds to each.

Event	System Response
Customer submits quotation form	System stores request and sends to employee
Employee fills quotation	System calculates total and saves as draft
Manager approves quotation	System changes status to "Approved"
Manager rejects quotation	System notifies employee
Approved quotation is ready	System emails it to customer automatically
Admin clicks "Generate Report"	System filters data and creates PDF or table
Customer submits feedback	System stores it in Feedback DB

Structure Chart

This shows how the modules in the system are organized, from top to bottom.



System Architecture

The system is designed as a **web-based client-server application** using modern technologies. All users access the system through a **web browser**, and the system runs on a **central server** connected to a **cloud database**.

Architecture Layers:

1. **Presentation Layer (Front-end)**
 - HTML, CSS, JavaScript
 - React.js or Vue.js
 - Runs in browser
 - Used by customer, employee, manager, admin
2. **Application Layer (Back-end)**
 - Node.js / Python / PHP
 - Controls logic, approvals, workflows
 - Connects frontend to database
3. **Database Layer**
 - MySQL or PostgreSQL
 - Stores customer info, quotations, reports, users, feedback
4. **Email and Notification Layer**
 - SMTP server for sending final quotations
 - Optional SMS/email APIs for alerts
5. **Security Layer**
 - HTTPS
 - Login authentication
 - Role-based access
 - Audit logs

Deployment

- Hosted on: Cloud (AWS, Azure, DigitalOcean)
- Database Backup: Daily automatic backup
- Access: Browser-based, secure login
- Support: Scalable, mobile-friendly

Summary

This physical DFD section describes how the new system will work in the real world. All users — customer, employee, manager, admin — will access the system online. Most tasks will be done automatically, and only a few will require human interaction. Data will be safe, fast to access, and well-organized.

The system architecture is designed to be **modern, secure, and easy to maintain**, while the structure is clear and simple for future developers to expand if needed.

8.0 System Wireframe (Input/Output Design)

This section explains how the **Quotation Management System (QMS)** will look and work from the user's point of view. It includes the **input designs** (what the user fills in or selects) and **output designs** (what the system shows or sends to the user).

Wireframes are simple drawings that show the **layout of each screen** before the actual system is built. These help designers and developers understand what each page should contain and how the users will interact with the system.

8.1 Input Design

Input design focuses on how the user gives information to the system. It must be:

- Easy to use
- Clear and simple
- Safe (with validation)
- Organized

1. Customer Quotation Request Form

Used by: Customer

Purpose: To send a new service quotation request

Input Fields:

- Full Name (Text)
- Email Address (Text)
- Phone Number (Text)
- Select Service Type (Dropdown: Housing, Education, Oil Refining, Industry)
- Description of Request (Textarea)
- Submit Button

2. Employee Quotation Entry Form

Used by: Employee

Purpose: To prepare and fill the quotation details

Input Fields:

- Select Customer (Dropdown with search)
- Service Description (Textarea)
- Price per Item (Number)
- Quantity (Number)
- Total Amount (Auto-calculated)
- Notes or Comments (Optional)
- Save Draft Button

3. Manager Review Panel

Used by: Manager

Purpose: To approve or reject a prepared quotation

Input Fields:

- Quotation ID (Auto)
- Customer Name (Read-only)
- Quotation Details (Read-only)
- Comment Box (Optional)
- Buttons: Approve / Reject

4. Admin Report Filter

Used by: Admin

Purpose: To view and generate reports

Input Fields:

- Select Time Period (Date Range Picker)
 - Select Department (Dropdown)
 - Status (All, Approved, Rejected, Pending)
 - Generate Report Button
-

8.2 Output Design

Output design shows how the system gives back information to the user. It must be:

- Easy to understand
- Neatly formatted
- Printable or downloadable when needed

1. Customer Confirmation Screen

Shown After: Submitting quotation request

Output Content:

- Thank you message
- Reference number for tracking
- Expected response time
- Email confirmation sent

2. Employee Dashboard

Shown After Login

Output Content:

- List of new and pending quotations
- Status of each quotation (Draft, Submitted, Approved, Rejected)
- Button to “Continue Draft” or “Create New”

3. Manager Dashboard

Shown After Login

Output Content:

- List of quotations waiting for approval
- Filters: By date, by employee
- Action: View, Approve, Reject

4. Final Quotation (PDF Output)

Sent to: Customer (via email)

Includes:

- Customer details
- Service summary
- Itemized pricing
- Total cost
- Terms and conditions
- Approved by manager signature or note

5. Reports Module Output

Used by: Admin

Output Types:

- Monthly report (number of quotations sent/approved)
 - Staff performance report (who completed how many)
 - Department summary report
 - Export as PDF or Excel
-

8.3 Wireframe Examples (Text Format)

Below are simple text wireframes. You can draw these using tools like draw.io, Figma, or even hand-sketch if needed.

A. Customer Request Form (Wireframe)

The wireframe shows a web form for requesting a quotation. At the top, there is a header with the logo 'DOSTTO' and the text 'Quotation Management System'. Below the header is a 'Back to Home' button. The main form is titled 'Request Quotation' and includes a sub-header: 'Please fill out the form below to request a quotation for our services. We'll get back to you within 3 working days.' The form contains several input fields: 'Full Name *', 'Email Address *', 'Phone Number *', 'Service Type *' (a dropdown menu), 'Description of Request' (a text area), 'Urgency Level' (a dropdown menu), and '\$ Budget Range (Optional)' (a dropdown menu). Below these fields is a section titled 'Additional Information' which contains a list of bullet points: 'What happens next?', 'We'll review your request within 24 hours', 'Our team will prepare a detailed quotation', 'You'll receive the quotation via email', and 'We're available for any questions or clarifications'. At the bottom of the form is a large blue button labeled 'Submit Request'. Below the button, there is a footer section with the text 'Need immediate assistance? Call us at +964 (0) 771 812 6767' and an email address 'Email: mkgmu202387@unq.edu.iq'. The footer also includes the copyright notice '© 2025 DOSTTO Quotation Management System. All rights reserved.'

DOSTTO Quotation Management System

[Back to Home](#)

Request Quotation

Please fill out the form below to request a quotation for our services. We'll get back to you within 3 working days.

Full Name *
Enter your full name

Email Address *
Enter your email address

Phone Number *
Enter your phone number

Service Type *
Select a service type

Description of Request
Please describe your requirements in detail...

Urgency Level
Normal (3-5 working days)

\$ Budget Range (Optional)
Select budget range

Additional Information

What happens next?

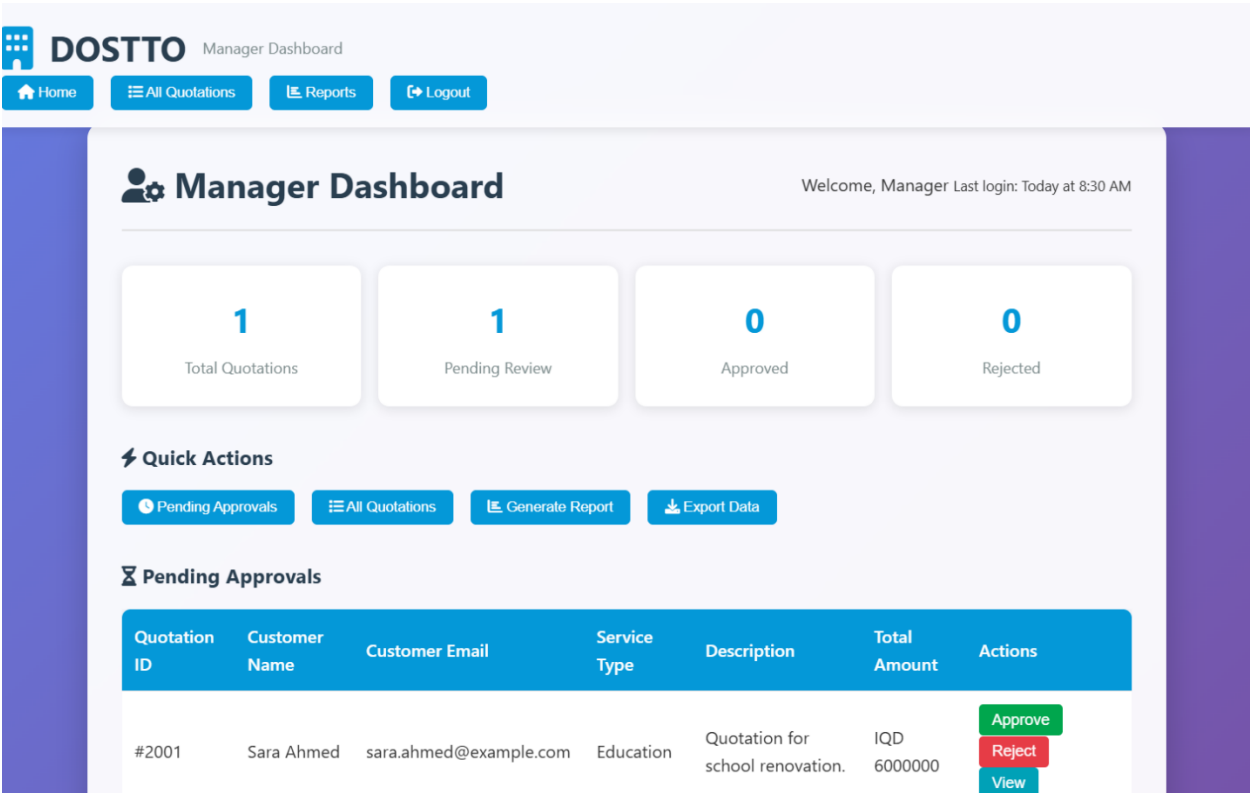
- We'll review your request within 24 hours
- Our team will prepare a detailed quotation
- You'll receive the quotation via email
- We're available for any questions or clarifications

[Submit Request](#)

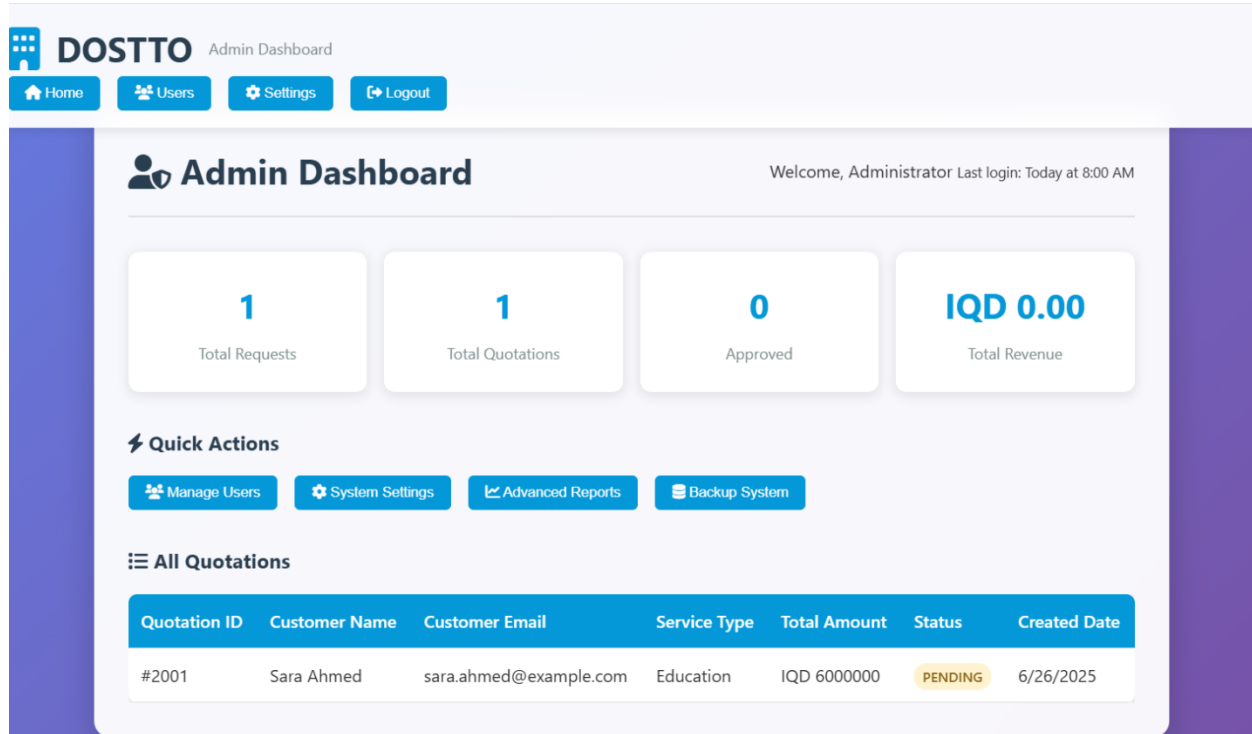
Need immediate assistance? Call us at +964 (0) 771 812 6767
Email: mkgmu202387@unq.edu.iq

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B. Manager Approval Screen (Wireframe)



C. Admin Report Screen (Wireframe)



Summary

The system wireframes show how each user will **interact with the system interface**. Inputs are simple and clear, while outputs are well-organized and meaningful. These wireframes help everyone understand how the final screens will look before programming begins.

Once the wireframes are approved, the development team can build a working prototype based on these layouts and designs.

9.0 Summary of the Proposed System

The proposed system is called the **Quotation Management System (QMS)**. It is a web-based software system developed for **DOSTTO**, a multi-sector company that works in Housing, Education, Oil Refining, and Industrial Services. This system is designed to solve the many problems found in the company's current manual process for handling quotations.

Purpose of the System

The main purpose of the system is to help DOSTTO:

- Handle customer quotation requests more quickly and accurately,
 - Keep all quotation data in one central place,
 - Reduce the use of paper and Excel files,
 - Allow managers to approve quotations easily,
 - Let admin staff generate useful reports,
 - Improve communication with customers.
-

Who Will Use the System

There are four main types of users:

1. **Customer** – Can request a quotation and give feedback.
2. **Employee** – Prepares the quotation and updates its status.
3. **Manager** – Approves or rejects quotations.
4. **Admin Staff** – Generates reports and manages users.

Each user will have their own login and only see the parts of the system they are allowed to use. This makes the system secure and easy to use.

Key Features of the System

Feature	Description
Customer Quotation Request	Customers can request services through an online form.
Quotation Preparation	Employees can prepare and edit quotations in a dashboard.
Quotation Approval	Managers can approve or reject quotations through a review panel.
Email Notifications	Final quotations are sent automatically to customers by email.
Report Generation	Admins can generate daily, weekly, and monthly reports with filters.
User Role Management	Admin can create, edit, or remove users and assign roles.
Feedback Collection	Customers can provide feedback on the service received.
Audit Logs	The system tracks actions by users to increase accountability.

Benefits of the System

The Quotation Management System brings many important benefits to DOSTTO:

1. Time Saving

- Quotations can be created, approved, and sent faster.
- No more need to search for old files or send emails back and forth.

2. Improved Accuracy

- Automatic price calculation reduces human errors.
- Data is entered directly into the system with clear validations.

3. Centralized Data

- All customer and quotation information is stored in one secure database.
- Admins and managers can find records quickly when needed.

4. Better Communication

- Managers receive real-time notifications about quotations waiting for review.
- Customers get clear updates and professional PDF quotations by email.

5. Security and Control

- Role-based login ensures that only authorized users can view or change data.
- Backup and encryption protect data from loss or unauthorized access.

6. Detailed Reports

- Admin can generate charts and summaries for tracking performance.
- Reports help management make better decisions.

7. Customer Satisfaction

- Customers receive faster service and professional responses.
- Feedback can be collected and used to improve the company's services.

System Technology and Architecture

- **Type:** Web-based application (can be accessed from any browser)
- **Technology Stack:**
 - Frontend: HTML, CSS, JavaScript (e.g., React.js)
 - Backend: Node.js / Python / PHP
 - Database: MySQL or PostgreSQL
 - Server: Cloud-based (AWS, DigitalOcean, etc.)
- **Access:** All users log in through a secure login system.
- **Security:** HTTPS, encrypted passwords, daily backups

How It Solves the Existing Problems

Existing Problem	How QMS Solves It
Manual data entry	Automated digital forms
Paper-based quotations	Online quotation builder
Lost files	Secure centralized database
Delayed approvals	Manager login with instant action
No reports	Built-in reporting module
Poor customer communication	Auto email with final quotation
Data duplication	One system controls all entries
Low accountability	User action tracking and audit logs

Future Scalability

The system is built to grow with the company. Later upgrades can include:

- SMS notifications
- Multi-language support
- Mobile app version
- Integration with accounting or invoicing software
- AI-powered quotation suggestions

Final Summary

The **Quotation Management System (QMS)** is a complete digital solution that meets the current and future needs of DOSTTO. It replaces slow, paper-based work with an organized and professional online platform. It helps employees do their jobs better, gives managers more control, and improves customer service.

With this system, DOSTTO will become more modern, more efficient, and more competitive in the market.